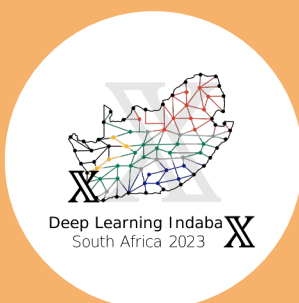


DEEP LEARNING INDABAX SOUTH AFRICA 2023

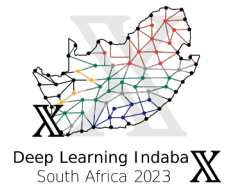


12 - 14 JULY 2023

Kramer Building,
University of Cape Town,
Cape Town, South Africa

DEEP LEARNING INDABA $\mathbb{X}$

SOUTH AFRICA 2023



ABOUT US

The Deep Learning Indaba $\mathbb{X}$ is a locally-organised meeting that helps ensure that knowledge and capacity in machine learning are spread more widely across the continent. The Indaba $\mathbb{X}$ is a means with which to experiment with the ways in which we strengthen African machine learning and its local communities, allowing more people to contribute to the conversation around machine learning and artificial intelligence. Indaba $\mathbb{X}$ events are local versions of the Deep Learning Indaba, which will be held this year in Accra, Ghana.

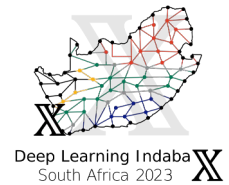
LIST OF ORGANISERS

Anna Bosman	Dinorego Bauba Mphogo	Thapelo Maupa
Christopher Currin	Guillaume Odendaal	Maryam Mohamad Al Mahdi
Maria Schuld	Nompilo Mkhulisi	Nomonde Khalo
Lydia de Lange	Avashlin Moodley	Yeshalen Naicker
Thapelo Sindane	Leo Hyams	Nilesh Jain
Nandi Mwase	Laing Lourens	Matimba Shingange
Chris Fourie	Joel Leonard	Claude Formanek
Aldrin Ngorima	Avashna Govender	Ruan de Kock
Callum Rhys Tilbury	Tharien Potgieter	

VENUE INFO

All talks take place at the **Kramer Building, University of Cape Town**:
Kramer Law Building, Wilfred and Jules Middle Campus of South Africa, Rondebosch, Cape Town, 7700

DEEP LEARNING INDABA SOUTH AFRICA 2023



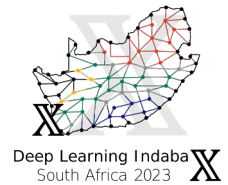
TRACKS

KEYNOTE		POSTER SESSION	
FOUNDATIONS		HACKATHON	
RESEARCH		WORKSHOPS	
APPLIED ML & INDUSTRY		PANEL: AI & AFRICA	

OVERVIEW

	11/07 TUESDAY	12/07 WEDNESDAY	13/07 THURSDAY	14/07 FRIDAY
8:00 am	Arrivals	Morning Coffee	Morning Coffee	Morning Coffee
8:30 am		Opening Remarks		
9:00 am				
9:30 am				
10:00 am		Break	Break	Break
10:30 am				
11:00 am				
11:30 am				
12:00 pm		Lunch	Lunch	Lunch
12:30 pm				
1:00 pm	Registration			Women in Tech Workshop AI Safety Workshop ML in Healthcare Workshop
1:30 pm				
2:00 pm		Break	Break	
2:30 pm				
3:00 pm				
3:30 pm				Break
4:00 pm		Group Photo & Speed Dating	Break	Panel: AI & Africa - A New Day or a Painful Replay?
4:30 pm				
5:00 pm		Opening Event: AI Fest & Poster session		Hackathon Prizes and Closing
5:30 pm				Travel
6:00 pm			Hackathon with Fruitpunch & Zindi	InstaDeep Movie Night
to late				

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 0

Tuesday, 11 July

ATTENDEE REGISTRATION

Kramer Building, University of Cape Town

13h00 - 15h30

**DAY 1
(Morning)**

Wednesday, 12 July

MORNING COFFEE

08h00 – 08h25

OPENING REMARKS

08h30 - 09h00

KEYNOTE

LT1

Paul Amayo - Building Intelligence for Robots in Africa

09h00 - 10h00

TEA/COFFEE BREAK

10h00 – 10h25

FOUNDATIONS

LT1

Daniel N, Wike - Decoding Data Science to Encode Concepts

10h30 - 12h00

RESEARCH

LT2

Aurora Geber - The Digital-first World

10h30 - 11h00

Isaac Itumeleng Setshedi - Unveiling Hidden Parameters: Transforming the Classic Solve for X Problem to Data Driven Interpretable Latent Parameters

11h00 - 11h30

Deshen Moodley - Embracing Human-Centred Artificial Intelligence: The Rise of Augmented AI Systems

11h30 - 12h00

APPLIED ML & INDUSTRY

LT3

Jacques Ludik - Pushing AI Innovation to Develop Trustworthy Explainable AI, AGI and Autonomous Intelligent Agents in a Decentralised World

10h30 - 11h00

Stephanie Muller & Gemma Dawson - What's Next at IBM Research & IBM Quantum?

11h00 - 11h30

Gabi Immelman - Minds and Machines: The Most Powerful Technology in the Classroom?

11h30 - 12h00

LUNCH

12h00 – 12h55

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 1
(Afternoon)

Wednesday, 12 July

KEYNOTE

LT1

Kshitij Thorat - Translating Astronomical Big Data into Patterns of Discovery

13h00 - 14h00

TEA/COFFEE BREAK

14h00 – 14h25

FOUNDATIONS

LT1

Ruan van der Merwe - Paying Attention to Transformers: A Deep Dive into Transformer Networks

14h30 - 16h00

RESEARCH

LT2

St John Grimby - Neuro-Symbolic Intro to Causality

14h30 - 15h00

Emile-Reyn Engelbrecht - Semi-Supervised Learning and Novelty Detection Using GANs

15h00 - 15h30

Cassandra Durr - Contrastive Learning Fundamentals and Practical Applications

15h30 - 16h00

APPLIED ML & INDUSTRY

LT3

Sedinam Simpson - A Day in Life of a Data Scientist

14h30 - 15h00

Karen Viljoen - Data Plumbing 101

15h00 - 15h30

Dineo Makoro - How a Line Graph can Ruin Your Day

15h30 - 16h00

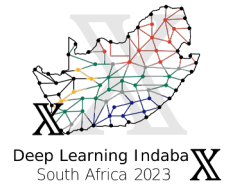
GROUP PHOTO & SPEED DATING

16h00 - 17h00
Outside

OPENING EVENT: AI FEST & POSTER SESSION

17h00 – 20h30
Kramer Quad

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 2
(Morning)

Thursday, 13 July

MORNING COFFEE

08h00 – 08h55

KEYNOTE

LT1

Jay van Zyl - The Human Organism as an Algorithm

09h00 - 10h00

TEA/COFFEE BREAK

10h00 – 10h25

FOUNDATIONS

LT1

Ritesh Kanjee - Mastering Modern Computer Vision Workshop From Basics to Cutting-Edge Technologies

10h30 - 12h00

RESEARCH

LT2

Herman Kamper - What can Large Spoken Language Models Tell Us About Speech?

10h30 - 11h00

Matthew Baas - Voice Conversion With Just Nearest Neighbours

11h00 - 11h30

Leanne Nortje - Visually Grounded Few-Shot Word Learning in Low-Resource Settings

11h30 - 12h00

APPLIED ML & INDUSTRY

LT3

Alta de Waal - AI Use Cases in BMW Group South Africa

10h30 - 11h00

Dr Frank Ortmann - MLOps in Enterprise

11h00 - 11h30

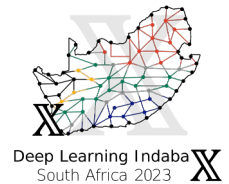
Shane Action - Natural Language AI is Revolutionising Medical Science

11h30 - 12h00

LUNCH

12h00 – 12h55

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 2
(Afternoon)

Thursday, 13 July

KEYNOTE

LT1

Emma Ruttkamp-Bloem - AI and Agency

13h00 - 14h00

TEA/COFFEE BREAK

14h00 - 14h25

FOUNDATIONS

LT1

Jacobie Mouton - An Introduction to Variational Inference and its Application in Deep Learning

14h30 - 16h00

RESEARCH

LT2

Mervyn Christoffels - High Performance Computing

14h30 - 15h00

Joseph Muthui Wacira - Advances in Machine Learning for Renewable Energy

15h00 - 15h30

Aby Louw - Text-to-Speech for African language (Resource-Scarce)

15h30 - 16h00

APPLIED ML & INDUSTRY

LT3

Keegan White - Smart Networks - the Power of Machine Learning Controlled Computer Networks

14h30 - 15h00

Thomas Gwasira - Reading the Air: Deep Learning in Radar, Communications and Acoustic Signal Processing

15h00 - 15h30

Mahmood-Ali Parker - Liveness Detection: Defending Against the Emerging Threat of Deepfakes

15h30 - 16h00

TEA/COFFEE BREAK

16h00 - 16h25

FOUNDATIONS

LT1

Asad Jeewa - Reinforcement Learning

16h30 - 18h00

LIGHTNING TALKS

LT2

16h30 - 18h00

HACKATHON PREPARATION

LT3

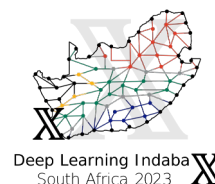
Mehrdad Ghaziasgar - An Unconvoluted Introduction to CNNs

16h30 - 18h00

HACKATHON WITH FRUITPUNCH & ZINDI

18h00 - 00h00
New Lecture Theatre
Upper Campus

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 3
(Morning)

Friday, 14 July

MORNING COFFEE

08h00 – 08h55

KEYNOTE

LT1

John Kamara - The AI Opportunity for Africa

09h00 - 10h00

TEA/COFFEE BREAK

10h00 – 10h25

FOUNDATIONS

LT1

Kevin Eloff - Denoising Diffusion Models: Introduction and Applications

10h30 - 12h00

RESEARCH

LT2

Sasha Abramowitz - Beyond Python: How Julia can Accelerate your Reinforcement Learning Projects

10h30 – 11h00

Batsi Ziki - Curiosity: The Compass that Leads to Intrinsic Motivation

11h00 – 11h30

Divanisha Patel - Reinforcement Learning and its Applications to Real World Problems

11h30 – 12h00

APPLIED ML & INDUSTRY

LT3

Dr Eileen Izette Carter - RightsTech Balancing Innovation and Human Rights in the Age of Deep Learning

10h30 - 11h00

Chijioke Okorie - Navigating Legal and Licensing Landscape for Machine Learning Applications

11h00 - 11h30

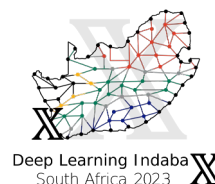
Roger Carthew - How People Power your Business

11h30 - 12h00

LUNCH

12h00 – 12h55

DEEP LEARNING INDABA SOUTH AFRICA 2023



DAY 3
(Afternoon)

Friday, 14 July

WORKSHOPS WOMEN IN TECH

13h00 – 15h30
LT1

Thenjiwe Kubheka - Developmental Opportunities for Women within the Technology Sector
Andrea Bohmert - Control Your Own Destiny
Mmaki Jantjies - Intentional Career development in the Field of AI
Panel discussion
Breakout sessions

AI SAFETY AND GOVERNANCE

LT2

Jonas Kgomo - A Guided Walkthrough of the African Safety Landscape
Zainab Taonga Chirwa - Perspectives on AI Governance in Africa
Benjamin Sturgeon - An Introduction to AI Safety

ML IN HEALTHCARE

LT3

Dr Oluwasegun Julius Aroba - Uncovering the Emergence of AI in Healthcare Industries in the 21st Century
Ayogebboh Epizitone - AI Applications in Healthcare
Siphiwe Thwala - Biomedical Text Insights Extraction
Kayode Adetunji - Using Deep Learning to Predict CVD Multimorbidity in Africa: A Guide to Navigate the Dark Abyss
Junior Muka - Improving Data Quality using Machine Learning
Panel discussion

TEA/COFFEE BREAK

15h30 – 15h55

PANEL DISCUSSION

AI & Africa: A New Day or a Painful Replay?

LT1

16h00 – 17h00

Panelists: Sandras Phiri, Zara Schroeder, Mark Gaffley, Thapelo Nthite
Host: Callum Tilbury

CLOSING CEREMONY

17h00 - 17h30

Prize Giving for Posters & Hackathon

TRAVEL TO TOWN

17h30 - 18h00

CLOSING EVENT

InstaDeep Movie Night
and/or
Explore Town

18h00 – 20h30
The Labia Theatre
18h00 – 23h00
Kloof Street

TRAVEL TO RESIDENCES

23h00

DEEP LEARNING INDABA SOUTH AFRICA 2023



Deep Learning Indaba
South Africa 2023

PARTNERS



DEEP LEARNING
INDABA



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CENTRE FOR ARTIFICIAL
INTELLIGENCE RESEARCH

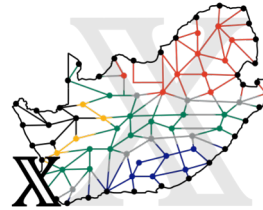


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